

Nopal-Sense

Mixed-Signal Soil Impedance Spectroscopy Hub

IEEE SSCS PICO Open-Source Chipathon 2026

Week 24 Project Proposal Review - June 12, 2026

Team members

Leader: Ernest Zermeño (@Jefemaestro33)

Individual Track B participant

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Track

B - Circuits for Sensors

Secondary relevance: D - AI/LLM-assisted audit and documentation workflow

Project Information

Goal

Nopal-Sense aims to tape out a low-power mixed-signal soil-impedance spectroscopy ASIC for agricultural deployments. The chip generates controlled multi-frequency excitation through soil electrodes, measures the root-zone response, and exposes calibrated data through a simple SPI-controlled hub.

High-Level Design

The differentiating block is the soil impedance spectroscopy path: DDS/DAC excitation, electrode interface, TIA/IQ readout, filtering, and shared SAR ADC. Around it, v1 adds SPI registers, scheduler, alerts, and bridge interfaces. Digital RTL is integrated with 185/185 Icarus assertions passing.

Application

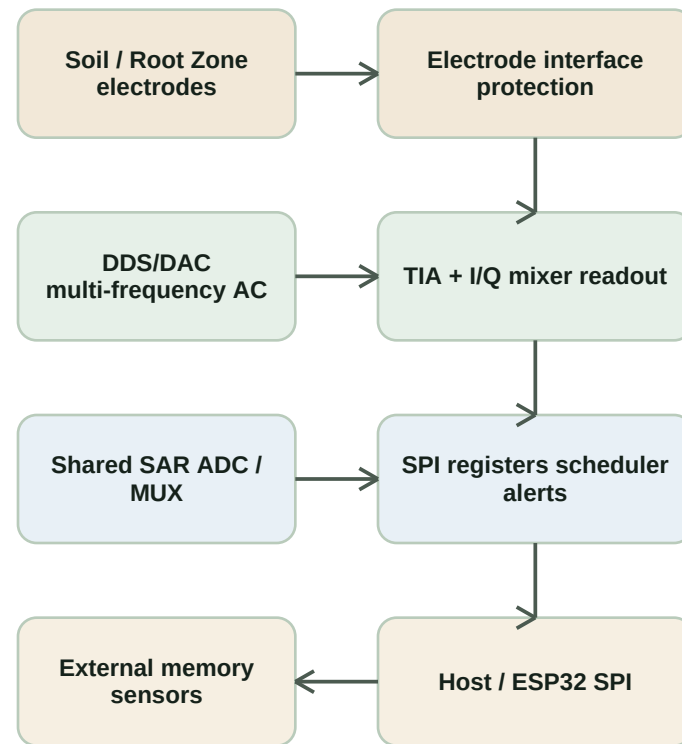
Low-cost agricultural soil impedance spectroscopy for irrigation, salinity, and root-zone tracking. Future validated datasets and software models can explore fungal/oomycete-related stress patterns.

References

SSCS Chipathon 2026 repo and guidelines; Nopal-Sense public repo/spec/architecture; Kelleners et al., 2009; Samouelian et al., 2005.

Nopal-Sense v1

soil impedance spectroscopy hub



Core claim: repeatable multi-frequency soil response data.

Team Background

Academic / Technical Experience

Ernest Zermeño - Biology background, currently working in bioinformatics research involving CRISPR and experimental biology. Technical interests include computational biology, data analysis, embedded systems, and measurement tools for biological and agricultural systems.

Work / Application Experience

Software and applied technology work through zlabstudio. Previous work includes exploring FHE-based software development and building practical software systems, with a focus on turning technical ideas into usable tools.

Why this project fits

The project is framed around practical soil/root-zone measurements where biology, software, data, and mixed-signal instrumentation overlap.

Questions, Suggestions, Doubts?

- 1 Is the v1 must-work scope realistic if limited to soil impedance spectroscopy AFE, shared SAR ADC, SPI register access, and low-power scheduling?
- 2 Should the ADC be a custom SAR implementation in v1, or should Nopal-Sense reuse an existing open-source/reference ADC to reduce schedule risk?
- 3 What electrode protection, ESD strategy, and analog pad usage are recommended for soil probes connected outside the chip package?
- 4 Are 10 kHz, 30 kHz, and 100 kHz acceptable first-tapeout frequencies for root-zone impedance, or should the top frequency be reduced for robustness?
- 5 What minimum AMS validation evidence is expected by the July block and top-level simulation reviews?

Requested mentor feedback: keep v1 conservative enough for first tapeout while preserving the soil spectroscopy differentiator.